

GEM Company

China

BATTERY TECHNOLOGY | EV SUPPLY CHAIN | MARKET EXPANSION

Acquisition Overview for Tesla Inc.

Strategic Consolidation of EV Manufacturing & Battery Technology

Prepared by INSEAD Group 32 on 12 July 2023

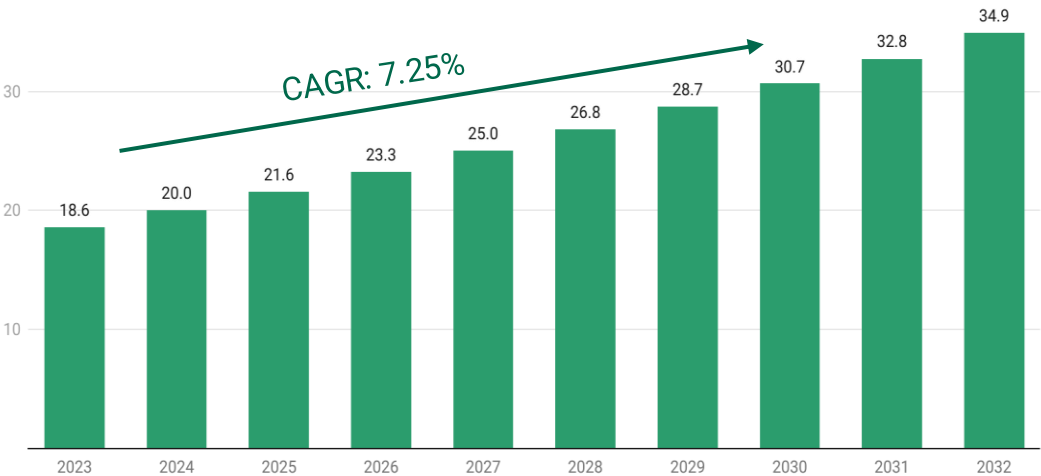
Battery Recycling Industry Overview and Market Trends

Major Segments (2023)

- Market segmentation by:
- Application: **automotive & transportation** (58%), industrial, consumer and electronic appliances, others
 - Battery type: Lead acid, Nickel, **Lithium-ion** (highest CAGR due to rise in EV), others
 - Region: **Asia Pacific** (biggest market share of 44%) and Europe (fastest growth)

Market Size (2023 - 2032)

Global Battery Recycling Market Size (\$ Bn)



Source: Mordor Intelligence, Spherical Insights • Created with Datawrapper

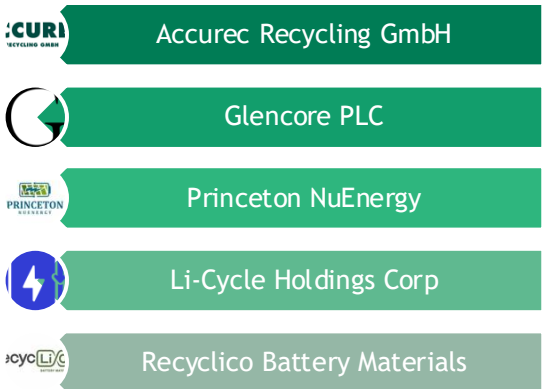
Key Facts

- Battery recycling market is expected to grow from USD 18.6 bn in 2023 to USD 34.9 bn in 2032 with a CAGR of 7.25%
- Lithium-ion battery expected to witness significant growth
- Asia Pacific has the highest industry market share of 44%. Meanwhile, Europe is the fastest-growing market during the forecast period due to the number of the rising usage in electric vehicles (EVs)

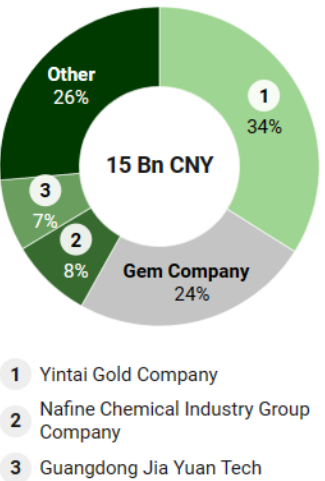
Market Share (2023)

Market concentration: relatively highly **fragmented** and competitive market without dominant players

Global Market Players



Market Capitalization Comparison with Chinese Industry Competitors

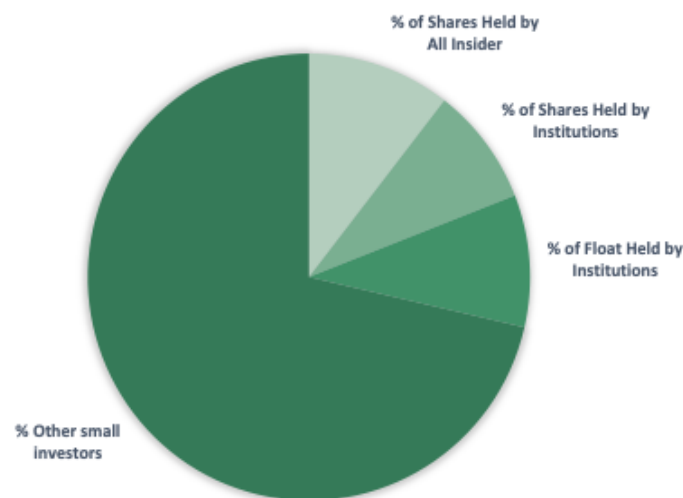


GEM Company Overview

Mission

- Promoting the philosophy of "limited resources, unlimited recycling":
 - Advocating for mining urban mines for pollution elimination and resource rebuilding
 - Leading enterprise in China's circular economy
 - Promoter of the global new energy industry

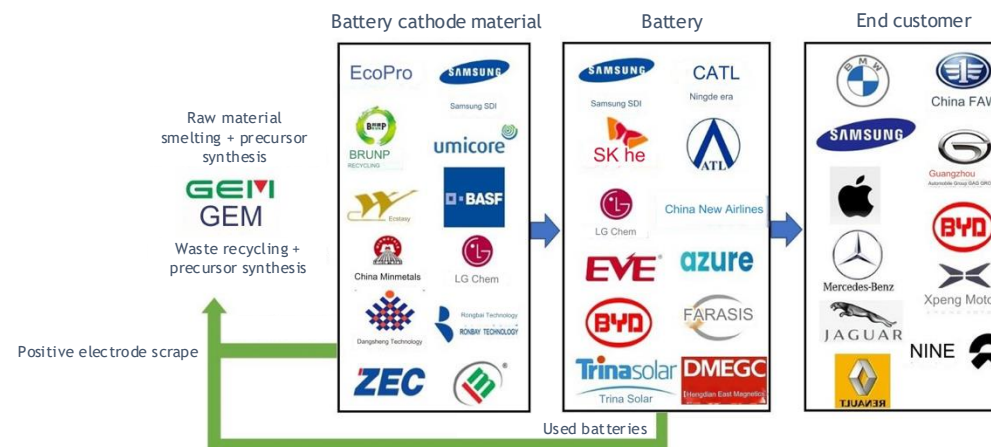
Ownership



Company description

- Company established in 2001
- Listed on the Shenzhen Stock Exchange (Stock Code: 002340) and SIX Swiss Exchange
- Recycling and regenerating over 30 scarce resources, including cobalt, nickel, lithium, copper, tungsten, gold, and rare earth.
- Principal business: dual tracks of "new energy materials manufacturing + urban mining"
 - manufacturers ternary precursors for power batteries, ternary cathode materials and tricobalt tetraoxide for 3C batteries
 - recycles and treats waste and regenerates scarce resources

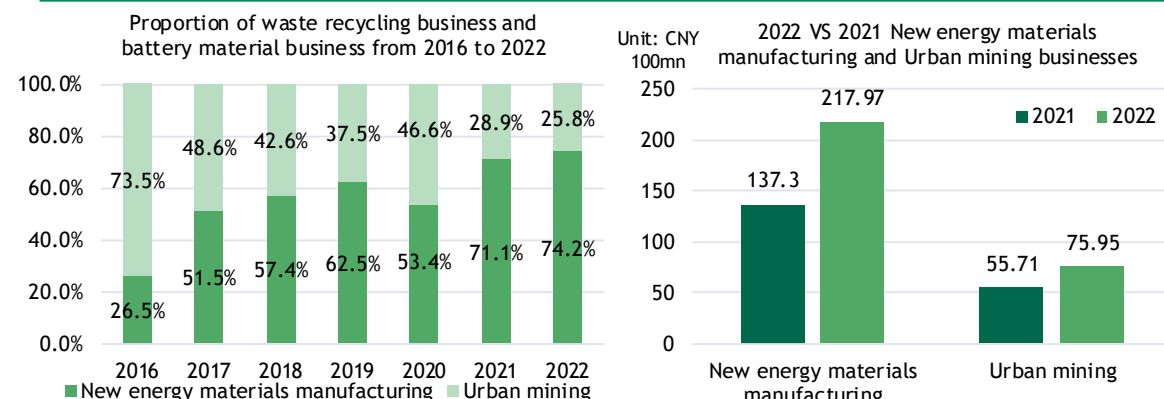
Key products and services



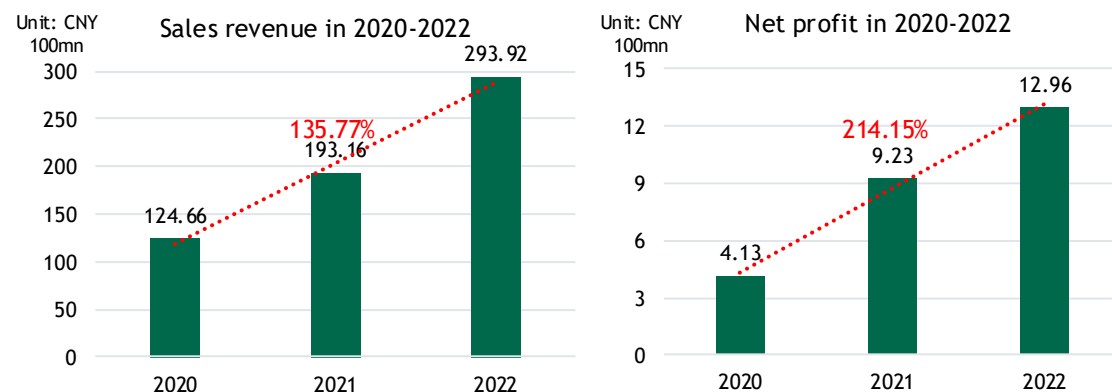
New Energy Materials Market Chain

GEM Company Overview

Business Scope



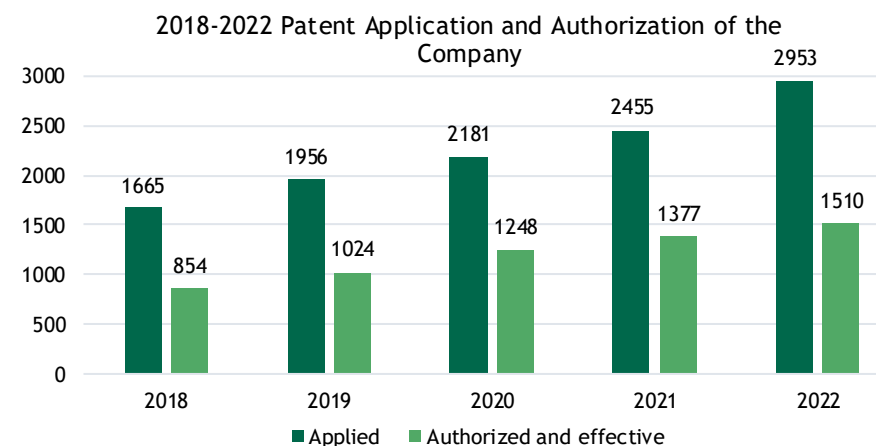
Growth



Value Chain Mode



Patent Hold

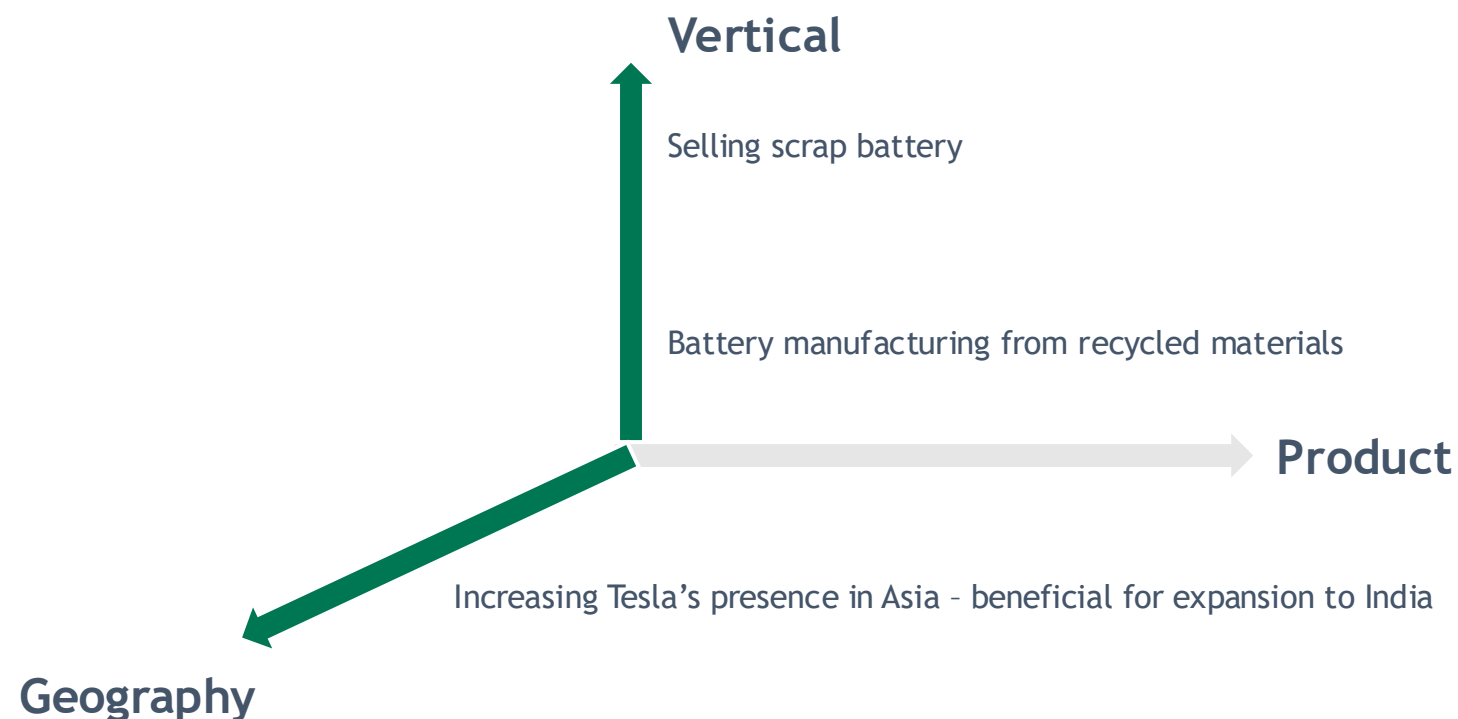


Deal Rationale

Proposed strategy

- Enhancing Tesla's corporate strategy by purchasing a **battery recycling** and **manufacturing** company that could be **vertically integrated** within Tesla's value chain
- It is recommended that the company is **absorbed** after the acquisition to reach cost synergies

Vertical integration



Deal Rationale

VRIO Framework

Resources	Valuable?	Rare?	Inimitable?	Organized?	Impact of Resource
Recycling technology know-how	YES	YES	YES	YES	Ability to recycle in-house
Logistics expertise	YES	YES	YES	YES	Ability to recycle in-house
Presence on Asian markets	YES	YES	YES	YES	Enhance expansion to Asian markets, mainly India
Battery technology know-how	YES	YES	NO	YES	Enhance current battery production capability

Deal Rationale

Recommended acquisition integration approach

Comments



- It is recommended that GEM Company is absorbed after the acquisition because:
 - In order to meet the cost synergies (economies of scale) full integration is vital
 - Based on research and previous Tesla’s experience cultural integration is possible

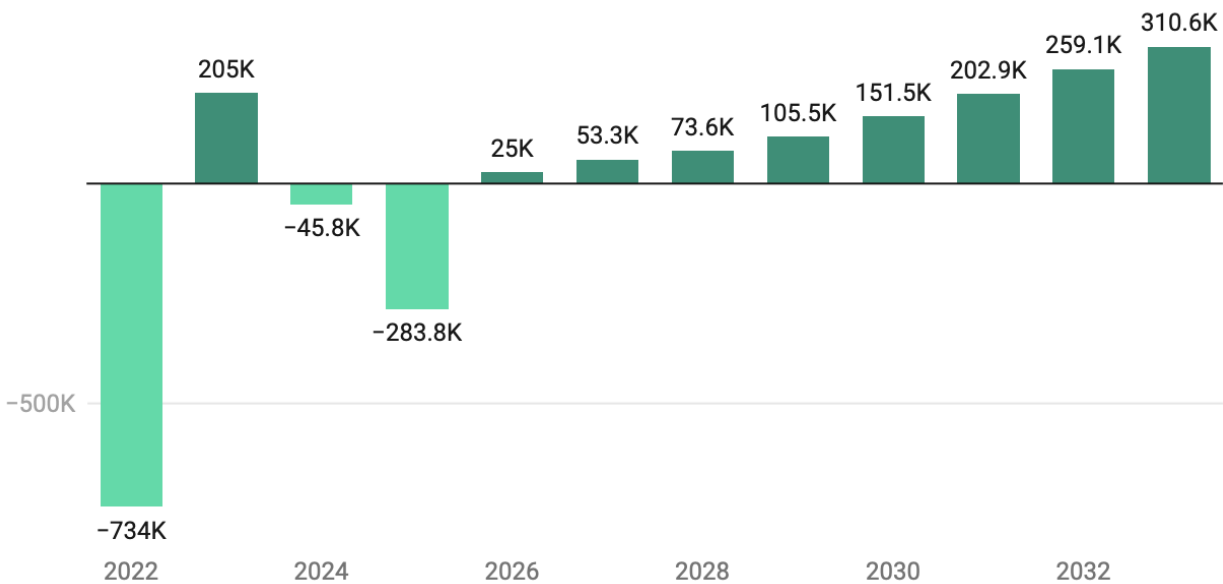
Valuation Consideration

Assumptions

Revenue growth: 16.1% (same as the 2025 forecast)
EBIDTA margin: gradual increase from 8.3% to 10% over 10 years
CAPEX: 5.2% (same as the 2025 forecast)

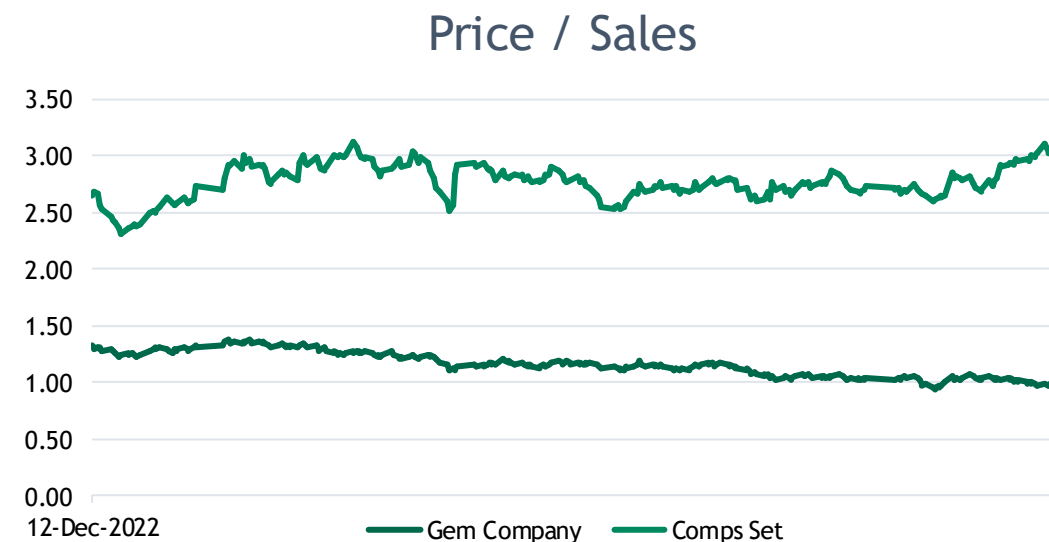
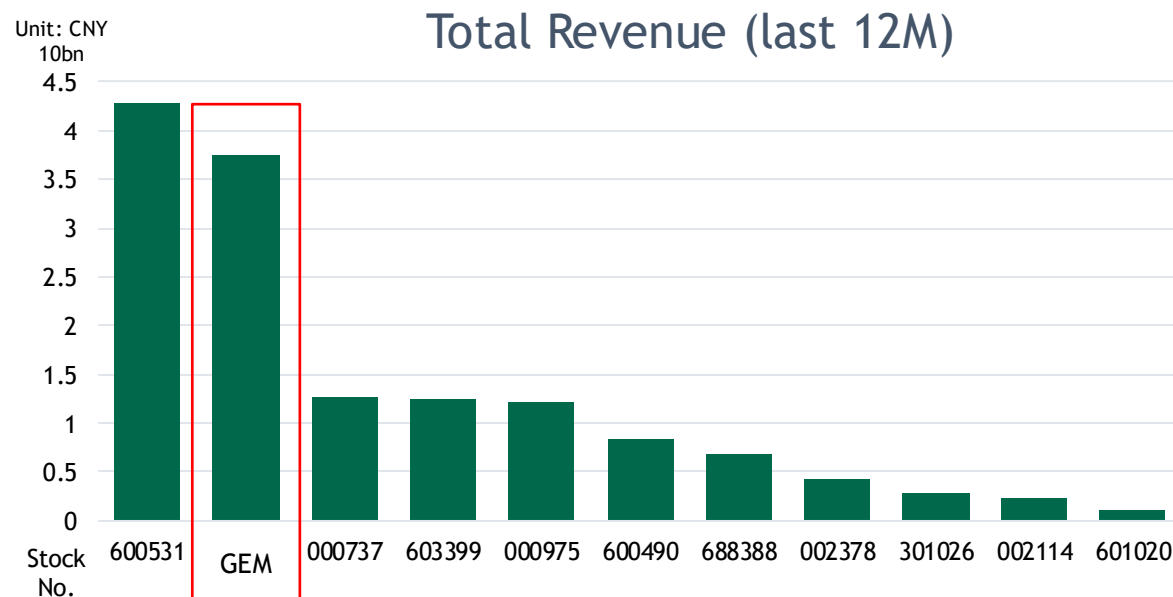
Tax: 13.8% (historical average)
Long Term Growth Rate: 3.5% (1.5% GDP growth 10 years, and 2% inflation rate in the USA)

Unlevered Free Cash Flows (thousands of euros)



WACC	6.4%
Terminal value: Growth in perpetuity approach	
Long term growth rate	3.5%
2033 FCF x (1+g)	310,645
Terminal value in 2033	10,620,921
Present value of terminal value	5,354,131
Present value of stage 1 cash flows	597,624
Total enterprise value (TEV)	5,951,756

Peer Valuation



GEM's total revenue is outperforming its competitors, but its price/Sales is way lower than the Competitors, putting Tesla in an advantageous position to acquire GEM.

Synergies

	2023	2024	2025	2026	2027	2028	2029	2030	2031
Cost synergies - Delta COGS		0.00	0.00	183.48	220.58	260.86	308.49	364.83	431.45
Discounted value		0.00	0.00	152.22	171.95	191.07	212.32	235.93	262.17

Discount factor = WACC

PV of synergies \$ 1225.651

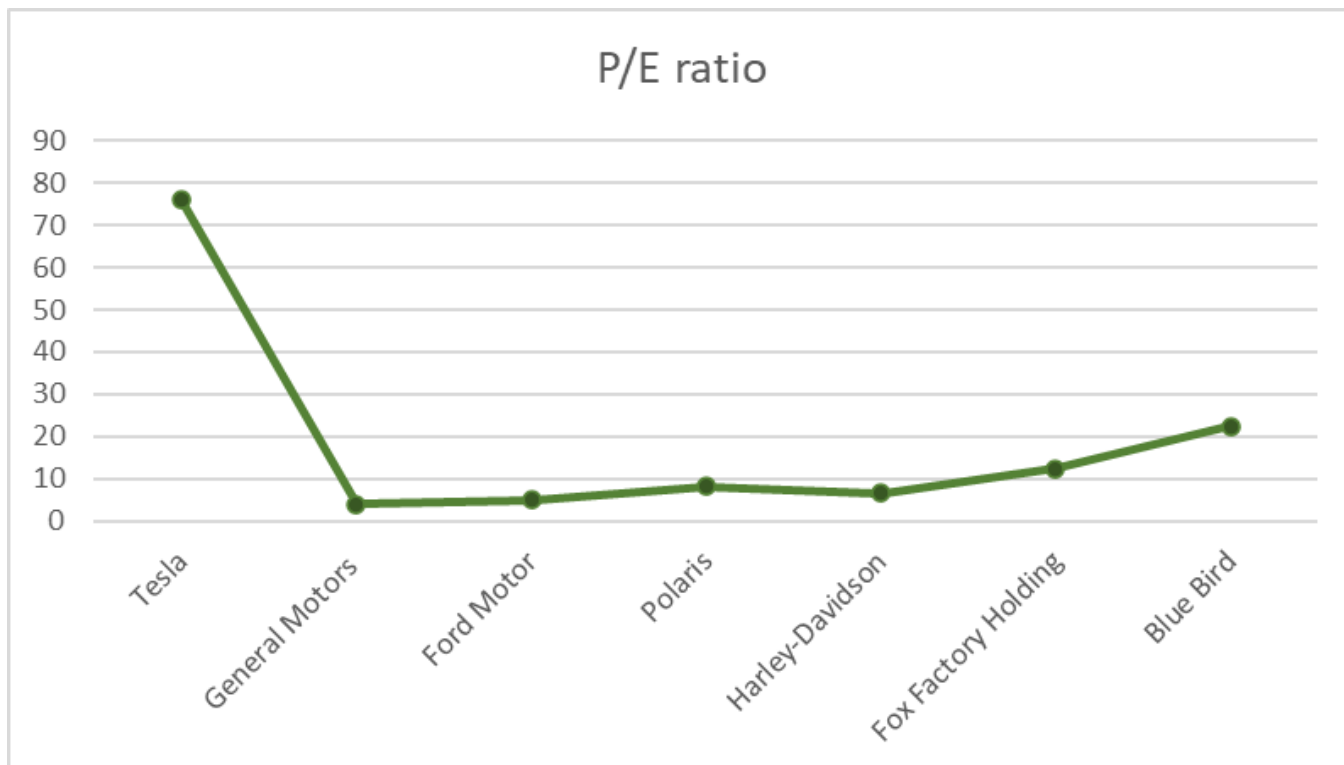
EUR/USD (13/12/2023) 1.08

PV of synergies € 1134.862

The cost synergies stop after 2031

- This acquisition will bring **cost synergies** to Tesla that will reduce its COGS by 0.1% per year through 2031
- Synergies will take place **from 2026 onwards**, according to PMI
- **After 2031**, there will be **no more synergies to realize**
- Through this merger, Tesla will achieve **vertical integration** both forward and backward

Tender Offer



Total Price = Equity Value (4.05bn EUR) + Synergies (1.13bn) * 0.75

Synergies are contingent and we pay them only after they materialize in the next 3 years

For the initial payment, we pay 80% in stock (since P/E is 76.21 compared to lower values for peers) and then the remaining payments are made after 2 years in the excess cash Tesla has (15bn USD in 2022).

Our Credentials

Antonio Cereda



Sarthak Goel



Adam Kubanek



Elisabeth Vandeurzen



Ausie Widawati



Yaou Zhang



Our past clients



Conclusion and Recommendation

Risks

- The market is heavily fragmented, with a lot of small layers and shared technologies
- Geographically, GEM is in China, which is outside of Tesla's geographical expertise

Recommendations

- Acquire GEM Company, a battery recycling and manufacturing company that could be **vertically integrated** within Tesla's value chain
- Initial payment of 80% in stock and 20% contingent pay in cash after 2 years
- **Absorption** after the acquisition to reach cost synergies

Next Steps

- Explore alternative ways of entering the battery recycling market and compare with the acquisition of GEM Company
- Make a tender offer to GEM Company and conduct comprehensive Due Diligence of the company